COUNTEREXAMPLES TO NAHM'S CONJECTURE ON THE BLOCH-GROUP TORSION CRITERION FOR MODULARITY

ARGUS OPEN RESEARCH PROJECT

Abstract. We record the counterexamples of Vlasenko and Zwegers (2011) showing that the q-hypergeometric Nahm series $f_{A,B,C}(\tau)$ can be modular for positive-definite rational $r \times r$ matrices A even when the associated element in the Bloch group fails the torsion condition of Nahm's Conjecture 4.1.

1. Main Refutation

Theorem 1.1 (Vlasenko–Zwegers, 2011)**.** *For rank $r \geq 2$, there exist explicit positive-definite symmetric matrices $A \in \mathrm{Mat}_r(\mathbb{Q})$, vectors $B \in \mathbb{Q}^r$, and scalars $C \in \mathbb{Q}$ such that the Nahm series*

$$f_{A,B,C}(\tau) \;=\; \sum_{n \in \mathbb{Z}^r_{\geq 0}} \frac{q^{\frac{1}{2} n^{\mathsf{T}} A n + B^{\mathsf{T}} n + C}}{(q)_n}$$

is a modular function, yet the real solution of the Nahm system $1 - z_i = \prod_{j=1}^{r} z_j^{A_{ij}}$ does not define a torsion element in the Bloch group $\mathcal{B}(\overline{\mathbb{Q}})$.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000731 in the domain of number theory, modular forms, and algebraic K-theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: necessity of the Bloch-group torsion condition for modularity in Nahm Conjecture 4.1 (Vlasenko-Zwegers 2011).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 20000731 (number theory, modular forms, and algebraic K-theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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